

Ryan Qin

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Education

University of Waterloo

Sept 2024 - Jun 2029

Bachelor of Applied Science in **Electrical Engineering**, Honours, Co-op

Technical Skills

Hardware/Embedded: FPGA, RTL, STM32, DSP, RISC-V, FSM, Robotics, RTOS, Linux, TCP/IP

Tools: Cadence, Altium, LtSpice, COMSOL, MATLAB, DSO, VNA, Vivado, Fusion 360, TensorFlow

Languages: C, C++, C#, Assembly, Verilog/SystemVerilog, VHDL, Java, Python, HTML/CSS, JavaScript

Professional Experiences

HSIO Compliance Engineer | TensorFlow, FPGA, C++, C#, RTOS, Cadence, DSO, MATLAB

May 2026 - Present

AMD

- Designing a **DisplayPort Waveform Decoder** in **MATLAB/C++**, aimed to debug **DP Link-Layer** directly from **oscilloscope captures** instead of a dedicated Link Layer Tester
- Developing a **STM32-based** fault-analysis **telemetry** using **supercapacitors** and **FreeRTOS**, helping to **recover** I/O, Power distribution, and thermal fault data during **brownouts** and **system hangs**
- Engineered a **TCP/IP-based PDU monitoring tool** using **Tensorflow**, designed to detect power draw anomalies for intermittent hangs or device stutters on **black-box DUTs** with **active-PFC supplies**
- Automated Display validation using **OpenCV**, **SSIM**, and **ROI-based image verification**, reducing debugging time by **30%**

Electrical Engineer | LtSpice, Altium, EPLAN, C++, RTOS, TCP/IP

Sep 2025 - Dec 2025

ATS Corporation

- Managed **full-cycle** design and verification for **four parallel** electronics projects, ensuring **on-time, spec-compliant** delivery
- Developed an **inductively isolated** pulse conditioner board in **Altium**, protecting PLC against stray voltage while **reducing 80% of Sub-GHz noise** from turbocharger's hall-effect tachometer
- Developed a **TCP/IP-enabled STM32** pneumatic leak-testing control board with a **thermal flow sensor** and **customized hardware drivers**, reducing power consumption by **40%** while decreasing unit's size by **50%**
- Designed and implemented **PID control loops** and **dither circuits** for a fuel nozzle testing system's pressure regulator, enabling **25%** more accurate simulation of turbojet engine operating conditions.

Electrical Coordinator | C#, .Net, COMSOL, MicroStation

Jan 2025 - May 2025

WSP

- Recognized by **WSP Canada's** Rail and Transit **VP** for OCS engineering Software's **innovation intake** potential
- Independently developed Overhead Contact System (**OCS**) engineering software in **C#**, reducing Calculation time by **40%**
- Optimized traction power substation's cooling system layout in **COMSOL FEA**, aimed to extend transformer life by **10 years**

Projects

TakeOne 🏆 | CAN-Bus, C++, Python

Finalist, Hack the North 2026 300+ Teams, 1,000+ Participants Sep 2026

- Designed I/O architectures using **isolated USB** and **CAN Bus**, ensuring reliable communication against ground lift/loop
- Architected the Robots **42V/12V Power Delivery Network**, ensuring stable and reliable drivetrain system
- Engineered **real-time PID feedback control** for **subject tracking** using vision-based error correction
- Implemented trajectory simulation for **AI-generated camera shots** in **MuJoCo**, validating robot motion before execution

TTL/FPGA Processor 🏆 | Verilog, Vivado, RTL, FSM, FPGA, DSP, TTL

Dec 2025 - Jan 2026

- Implemented **pipelined vector/tensor operations** in Verilog, using **FPGA DSP slices** to increase throughput per clock cycle
- Built a **custom breadboard CPU**, implementing the ALU, register file, control unit, and clocking from **first principles**
- Extended the CPU in **Verilog** for **FPGA deployment**, adding **pipelining**, forwarding, memory mapped I/O, and **SPI flash**

Autodump 🏆 | C++, TCP/IP, Fusion 360, Python, OpenCV

3rd place at MakeCU Nov 2025

- Designed **ArUco** marker navigation system in **OpenCV**, allowing the robot to **autonomously navigate** back to home position
- Engineered an Arduino-based serial **distributed I/O** to reduce **PWM-signal jitter** across **servos** and **ultrasonic sensors**
- Architected the robot's **mechanical layout in Fusion 360** for rapid prototyping and reduced material waste.

Competition RC 🏆 | PyTorch, Fusion 360, LTSpice, MCU, AVR Assembly

Finalist at Formula Null May 2025

- Designed a **MOSFET-based airsoft motor ESC**, achieving a **fast lap record** of **17 seconds** compared to a **25s average**
- Implemented a **lane-keeping algorithm**, combining **PID** and **PyTorch-based** vision to adjust steering from USB camera
- Built an RC car with **50%** recycled material and custom **3D printed** steering assembly